Seawater Density Sensitivity to Temperature Using the TEOS-10 Gibbs SeaWater Oceanographic Toolbox

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Abstract

This study examines the sensitivity of seawater density to a 1,° C temperature perturbation using the Thermodynamic Equation of Seawater 2010 (TEOS-10). The calculation uses the Gibbs SeaWater (GSW) Oceanographic Toolbox directly from its C source files, compiled and linked with a C++ application.

The input observations consist of in-situ temperature, Practical Salinity, and depth. These observations are converted to sea pressure, Absolute Salinity, and Conservative Temperature using TEOS-10 algorithms. Seawater density is subsequently calculated for the original state and for a perturbed state in which temperature is increased by 1,° C.

The resulting density perturbation is calculated as the difference between the perturbed and original densities. For the profiles examined here, the temperature perturbation produces a decrease in seawater density throughout the water column, with the magnitude of the response varying as a function of depth and the thermodynamic state of the seawater.

1 Objective

The objective of this calculation is to quantify the local thermodynamic sensitivity of seawater density to temperature.

For each observation, the initial seawater state is represented by

$$(S_P, , t, , z)\,, \tag{1}$$

where

$$S_P = \text{Practical Salinity},\ t \qquad = \text{in-situ temperature},\ z = \text{vertical position}. \tag{2}$$

Depth is represented as a positive quantity, while TEOS-10 uses height above the sea surface. Therefore,

$$z = -d, \tag{3}$$

where d is the measured depth below the sea surface.

The temperature perturbation is defined as

$$\Delta t = 1,^{\circ}\,\mathrm{C}. \tag{4}$$

The perturbed temperature is consequently

$$t' = t + \Delta t. \tag{5}$$

For this experiment,

$$\boxed{t' = t + 1,^\circ \mathrm{C}} \tag{6}$$

while salinity and pressure are held at their observed values.
The density perturbation is then defined as

$$\boxed{\Delta\rho = \rho' - \rho} \tag{7}$$

where ρ is the density of the original seawater state and ρ' is the density after the temperature perturbation.

2 TEOS-10 Thermodynamic Framework

The Thermodynamic Equation of Seawater 2010 (TEOS-10) describes the thermodynamic properties of seawater using Absolute Salinity, Conservative Temperature, and pressure.

The principal thermodynamic state variables used in this calculation are therefore

$$(S_A, , CT, , p)\,, \tag{8}$$

where

$$S_A = \text{Absolute Salinity},\ CT = \text{Conservative Temperature},\ p = \text{sea pressure}. \tag{9}$$

Seawater density is represented as

$$\rho = \rho(S_A, CT, p). \tag{10}$$

Equivalently, density may be expressed through specific volume v:

$$\boxed{\rho = \frac{1}{v}} \tag{11}$$

with

$$v = v(S_A, CT, p). \tag{12}$$

Thus,

$$\boxed{\rho = \frac{1}{v(S_A, CT, p)}} \tag{13}$$

provides the fundamental thermodynamic relationship used by the density calculation.

3 Computational Sequence

The complete calculation can be summarized as

$$\boxed{(S_P, t, d, \phi, \lambda) \longrightarrow (p, S_A, CT) \longrightarrow \rho} \tag{14}$$

where ϕ and λ denote latitude and longitude.

The perturbed calculation follows

$$\boxed{(S_P, t + \Delta t, d, \phi, \lambda) \longrightarrow (p, S_A, CT') \longrightarrow \rho'} \tag{15}$$

with

$$\Delta t = 1,^\circ \text{C}. \tag{16}$$

The final quantity of interest is

$$\boxed{\Delta\rho\rho(S_A, CT', p)\rho(S_A, CT, p)} \tag{17}$$

where

$$CT' = CT(S_A, t + \Delta t, p). \tag{18}$$

4 Pressure From Depth

The GSW function `gsw-p-from-z` is used to convert vertical position into sea pressure.

The input depth is converted to TEOS-10 height according to

$$z = -d. \tag{19}$$

The pressure is then evaluated as

$$p = p(z, \phi, H_{\text{dyn}}), \tag{20}$$

where

$$z = \text{height relative to the sea surface, } \phi = \text{latitude, } H_{\text{dyn}} = \text{dynamic height contribution.} \tag{21}$$

For the present calculation, the dynamic-height contribution is taken according to the assumptions used by the implementation.

Pressure is reported in decibars:

$$p \quad [\text{dbar}]. \tag{22}$$

Pressure is not assumed to be exactly proportional to depth. The conversion depends upon gravitational acceleration and therefore includes a latitude dependence.

5 Practical Salinity and Absolute Salinity

The observations provide Practical Salinity, S_P. TEOS-10 uses Absolute Salinity, S_A, as its thermodynamic salinity variable.

The conversion is represented by

$$\boxed{S_A = S_A(S_P, p, \phi, \lambda)} \tag{23}$$

and is implemented using the GSW function
gsw-SA-from-SP .
Absolute Salinity is expressed in

$$[\mathrm{g}, \mathrm{kg}^{-1}]. \tag{24}$$

The distinction between S_P and S_A is important because the TEOS-10 density calculation is based on Absolute Salinity rather than Practical Salinity.

6 Conservative Temperature

The measured temperature is the in-situ temperature t.

TEOS-10 converts in-situ temperature to Conservative Temperature according to

$$\boxed{CT = CT(S_A, t, p)} \tag{25}$$

using the GSW function
gsw-CT-from-t .
For the perturbed state,

$$t' = t + \Delta t, \tag{26}$$

and therefore

$$\boxed{CT' = CT(S_A, t + \Delta t, p)} \tag{27}$$

with

$$\Delta t = 1,^\circ \mathrm{C}. \tag{28}$$

An important feature of this calculation is that the perturbation is applied to the original in-situ temperature before the TEOS-10 conversion to Conservative Temperature. Thus, the calculation does not simply assume

$$CT' = CT + 1,^\circ \mathrm{C}. \tag{29}$$

Instead,

$$CT' CT(S_A, t + 1,^\circ \mathrm{C}, p). \tag{30}$$

7 Density Calculation

Once Absolute Salinity, Conservative Temperature, and pressure have been determined, seawater density is calculated from

$$\boxed{\rho = \rho(S_A, CT, p)} \tag{31}$$

and the perturbed density is

$$\boxed{\rho' = \rho(S_A, CT', p).} \tag{32}$$

The density response is consequently

$$\boxed{\Delta\rho\rho(S_A, CT', p)\rho(S_A, CT, p)} \tag{33}$$

with units

$$[\Delta\rho]\mathrm{kg}, \mathrm{m}^{-3}. \tag{34}$$

8 Local Temperature Sensitivity

For a sufficiently small temperature perturbation, density may be approximated using a first-order Taylor expansion:

$$\rho(T + \Delta T) \approx \rho(T) + \left(\frac{\partial\rho}{\partial T}\right)_{S_A,p} \Delta T. \tag{35}$$

Therefore,

$$\boxed{\Delta\rho \approx \left(\frac{\partial\rho}{\partial T}\right)_{S_A,p} \Delta T} \tag{36}$$

where the derivative is evaluated at constant Absolute Salinity and pressure.
The thermal expansion coefficient α is defined by

$$\boxed{\alpha - \frac{1}{\rho}\left(\frac{\partial\rho}{\partial T}\right)_{S_A,p}} \tag{37}$$

and therefore

$$\left(\frac{\partial\rho}{\partial T}\right)_{S_A,p} - \rho\alpha. \tag{38}$$

Substitution gives

$$\boxed{\Delta\rho \approx -\rho\alpha\Delta T} \tag{39}$$

and, for the present experiment,

$$\boxed{\Delta\rho \approx -\rho\alpha} \tag{40}$$

because

$$\Delta T = 1,^\circ\,\mathrm{C}. \tag{41}$$

The actual calculation does not rely on this linear approximation. Instead, the complete TEOS-10 state is evaluated before and after the finite temperature perturbation.

9 Numerical Results

The generated CSV file contains the following variables:

Variable	Description
`time`	Observation time
`depth_m`	Depth below the sea surface
`temperature_C`	Original in-situ temperature
`perturbed_temperature_C`	Temperature plus 1°C
`SP_practical`	Practical Salinity
`SA_absolute_gkg`	Absolute Salinity
`pressure_dbar`	Sea pressure
`CT_C`	Original Conservative Temperature
`perturbed_CT_C`	Perturbed Conservative Temperature
`density_kg_m3`	Original density
`perturbed_density_kg_m3`	Perturbed density
`density_difference_kg_m3`	$\rho' - \rho$

Representative values from the generated profile are shown in Table **??**.

Table 1: Representative TEOS-10 results from the generated profile.

Depth (m)	t (°C)	S_A ($\mathrm{g\,kg^{-1}}$)	ρ ($\mathrm{kg,m^{-3}}$)	$\Delta\rho$ ($\mathrm{kg,m^{-3}}$)
10	17.976	33.630	1024.1565	-0.2484
20	17.770	33.635	1024.2551	-0.2468
61	12.389	33.181	1025.2615	-0.1968
145	8.992	33.861	1026.7693	-0.1657
302	7.154	34.185	1028.0231	-0.1504
499	5.796	34.354	1029.2534	-0.1406
754	4.769	34.541	1030.7134	-0.1358
1001	3.974	34.634	1032.0298	-0.1332
2213	1.955	34.796	1037.9917	-0.1416
3873	1.509	34.863	1045.6019	-0.1767

For example, at a depth of approximately 10 m, the calculated density is

$$\rho 1024.15649622\ \mathrm{kg,m^{-3}}. \tag{42}$$

After increasing the temperature by 1,° C,

$$\rho' 1023.90806198\ \mathrm{kg,m^{-3}}. \tag{43}$$

The resulting density difference is

$$\Delta\rho = \rho' - \rho \qquad = 1023.908061981024.15649622 \; = -0.24843424 \text{ kg}, \text{m}^{-3}. \tag{44}$$

Thus,

$$\boxed{\Delta\rho - 0.24843424 \text{ kg}, \text{m}^{-3}} \tag{45}$$

for this particular observation.

10 Depth Dependence

The magnitude of the density response changes with depth.

Near the surface, the profile produces density changes of approximately

$$\Delta\rho \approx -0.25 \text{ kg}, \text{m}^{-3} \tag{46}$$

for a 1,° C temperature increase.

Near 1000 m,

$$\Delta\rho \approx -0.13 \text{ kg}, \text{m}^{-3}. \tag{47}$$

At approximately 3873 m,

$$\Delta\rho \approx -0.177 \text{ kg}, \text{m}^{-3}. \tag{48}$$

This variation demonstrates that the temperature-density relationship is not represented adequately by a single constant coefficient over the entire water column.

The thermodynamic state of the seawater, including pressure, temperature, and salinity, changes substantially with depth.

11 Geographic Dependence

Latitude is required by the TEOS-10 pressure calculation and may also affect the conversion between Practical and Absolute Salinity.

The pressure calculation can therefore be represented schematically as

$$p = p(z, \phi, H_{\text{dyn}}). \tag{49}$$

The Absolute Salinity calculation depends upon geographic position:

$$S_A S_A(S_P, p, \phi, \lambda). \tag{50}$$

Consequently, latitude and longitude are part of the thermodynamic calculation even though they are not themselves thermodynamic state variables.

The current CSV output does not contain latitude or longitude fields. Therefore, the CSV alone cannot be used to establish the geographic location of the profile or its distance from the United States coast.

For complete reproducibility, the output should include

$$\phi = \text{latitude} \tag{51}$$

and

$$\lambda = \text{longitude}. \tag{52}$$

These quantities should preferably be recorded in the CSV alongside each profile or in associated metadata.

12 Software Implementation

The calculation is implemented as a C++ program linked directly against the required GSW C source files.

The principal source files are

- `cce1_teos10_density.cpp`
- `gsw_oceanographic_toolbox.c`
- `gsw_saar.c`

along with the associated TEOS-10 header and data files.

The C source files are compiled into object files and linked directly with the C++ application.

A representative compilation sequence is

```
gcc -c -O2 gsw_oceanographic_toolbox.c
gcc -c -O2 gsw_saar.c

g++ -O2 cce1_teos10_density.cpp
gsw_oceanographic_toolbox.o
gsw_saar.o
-o cce1_teos10_density
```

The resulting executable produces

```
cce1_teos10_density.csv
```

containing the original and perturbed thermodynamic states.

13 Interpretation

The results demonstrate that increasing seawater temperature by 1,° C decreases seawater density for the states represented by the profile.

The response is not constant with depth because seawater density is a function of multiple thermodynamic variables:

$$\rho = \rho(S_A, CT, p). \tag{53}$$

The experiment isolates the temperature component by holding the observed salinity and pressure fixed while modifying temperature.

Thus, the experiment should be interpreted as a controlled thermodynamic sensitivity calculation rather than as a complete model of future ocean warming.

In particular, it does not model simultaneous changes in salinity, mixing, circulation, evaporation, precipitation, air-sea heat flux, or water-mass formation.

14 Limitations

The principal limitation is that the experiment applies a prescribed temperature perturbation while holding other measured quantities fixed.

The calculation therefore evaluates

$$\rho(S_A, CT + \Delta CT, p)\rho(S_A, CT, p) \tag{54}$$

under the prescribed perturbation rather than simulating the full physical evolution of the ocean.

A further limitation is that the current CSV file does not contain latitude and longitude. These values should be retained as metadata for future analyses because they affect the pressure and Absolute Salinity calculations.

15 References

1. International Oceanographic Commission, Scientific Committee on Oceanic Research, and International Association for the Physical Sciences of the Oceans. *The International Thermodynamic Equation of Seawater—2010: Calculation and Use of Thermodynamic Properties*. TEOS-10 Manual, UNESCO, 2010.

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